

MAT 3150 — Homework 4

Due October 1

Problem 1 (10 points)

Let $(x_n)_n$ and $(y_n)_n$ be two sequences of real numbers, so that $(y_n)_n$ is increasing. Assume that

$$\lim_{n \rightarrow \infty} \frac{x_{n+1} - x_n}{y_{n+1} - y_n} = L,$$

for some $L \in \mathbb{R}$. Prove that for any $\varepsilon > 0$ there exists $N > 0$ so that

$$(L - \varepsilon)(y_{n+1} - y_N) < x_{n+1} - x_N < (L + \varepsilon)(y_{n+1} - y_N),$$

for all $n > N$.

Hint: Use $x_{n+1} - x_N = (x_{n+1} - x_n) + (x_n - x_{n-1}) + \cdots + (x_{N+1} - x_N)$, for any $n > N$.

Proof. Let $\varepsilon > 0$. For each k , write

$$r_k = \frac{x_{k+1} - x_k}{y_{k+1} - y_k},$$

so the hypothesis says $\lim_{k \rightarrow \infty} r_k = L$.

Step 1: Use the hypothesis. By the definition of $\lim_{k \rightarrow \infty} r_k = L$, there exists N such that for all $k \geq N$,

$$|r_k - L| < \varepsilon.$$

Unpacking the absolute value, $-\varepsilon < r_k - L < \varepsilon$, and adding L throughout gives, for all $k \geq N$,

$$L - \varepsilon < \frac{x_{k+1} - x_k}{y_{k+1} - y_k} < L + \varepsilon.$$

Step 2: Clear the denominator. We multiply the inequality from Step 1 by $y_{k+1} - y_k$. Since $y_{k+1} - y_k > 0$, multiplying by it preserves the direction of the strict inequalities $<$.

Indeed, since (y_n) is increasing, $y_{k+1} - y_k \geq 0$; and $y_{k+1} - y_k \neq 0$, since otherwise the fraction in the hypothesis would be undefined. Hence $y_{k+1} - y_k > 0$, and we obtain, for all $k \geq N$,

$$(L - \varepsilon)(y_{k+1} - y_k) < x_{k+1} - x_k < (L + \varepsilon)(y_{k+1} - y_k).$$

Step 3: Sum from $k = N$ to $k = n$. Let $n > N$. The inequality from Step 2 holds for each $k = N, N + 1, \dots, n$. Adding these $n - N + 1$ inequalities gives

$$\sum_{k=N}^n (L - \varepsilon)(y_{k+1} - y_k) < \sum_{k=N}^n (x_{k+1} - x_k) < \sum_{k=N}^n (L + \varepsilon)(y_{k+1} - y_k).$$

Since $L - \varepsilon$ and $L + \varepsilon$ do not depend on k , we can pull them out of the sums:

$$(L - \varepsilon) \sum_{k=N}^n (y_{k+1} - y_k) < \sum_{k=N}^n (x_{k+1} - x_k) < (L + \varepsilon) \sum_{k=N}^n (y_{k+1} - y_k).$$

Step 4: Simplify. The y -sum telescopes, since every $y_{N+1}, \dots, y_n$ appears once with $+$ and once with $-$:

$$\sum_{k=N}^n (y_{k+1} - y_k) = y_{n+1} - y_N.$$

Likewise, the x -sum telescopes (this is the hint):

$$\sum_{k=N}^n (x_{k+1} - x_k) = x_{n+1} - x_N.$$

Substituting both sums into the inequality from Step 3 gives

$$(L - \varepsilon)(y_{n+1} - y_N) < x_{n+1} - x_N < (L + \varepsilon)(y_{n+1} - y_N)$$

for all $n > N$, which is what we wanted to prove. □

Problem 2 (10 points)

Let $(x_n)_n$ and $(y_n)_n$ be two sequences of real numbers, so that $(y_n)_n$ is increasing and $\lim_{n \rightarrow \infty} y_n = \infty$. Assume that

$$\lim_{n \rightarrow \infty} \frac{x_{n+1} - x_n}{y_{n+1} - y_n} = L,$$

for some $L \in \mathbb{R}$. Prove that

$$\lim_{n \rightarrow \infty} \frac{x_n}{y_n}$$

exists and is also equal to L .

Hint: Use Problem 1.

Proof. Let $\varepsilon > 0$, and let N be as in Problem 1.

Step 1: Rearrange Problem 1's inequality. By Problem 1, for all $n > N$,

$$(L - \varepsilon)(y_{n+1} - y_N) < x_{n+1} - x_N < (L + \varepsilon)(y_{n+1} - y_N).$$

Adding x_N throughout,

$$x_N + (L - \varepsilon)(y_{n+1} - y_N) < x_{n+1} < x_N + (L + \varepsilon)(y_{n+1} - y_N).$$

Expanding the left half,

$$x_N + (L - \varepsilon)y_{n+1} - (L - \varepsilon)y_N < x_{n+1}.$$

and the right half,

$$x_{n+1} < x_N + (L + \varepsilon)y_{n+1} - (L + \varepsilon)y_N.$$

Since $y_n \rightarrow \infty$, there exists N_0 such that $y_{n+1} > 0$ for all $n \geq N_0$; for such n we may divide by y_{n+1} without changing the direction of the inequalities.

Dividing the left half by $y_{n+1} > 0$,

$$L - \varepsilon + \frac{x_N - (L - \varepsilon)y_N}{y_{n+1}} < \frac{x_{n+1}}{y_{n+1}}.$$

and the right half,

$$\frac{x_{n+1}}{y_{n+1}} < L + \varepsilon + \frac{x_N - (L + \varepsilon)y_N}{y_{n+1}}.$$

Together,

$$L - \varepsilon + \frac{x_N - (L - \varepsilon)y_N}{y_{n+1}} < \frac{x_{n+1}}{y_{n+1}} < L + \varepsilon + \frac{x_N - (L + \varepsilon)y_N}{y_{n+1}}.$$

Step 2: Handle the terms involving N . The leftover terms are $\frac{A}{y_{n+1}}$ and $\frac{B}{y_{n+1}}$, where

$$A = x_N - (L - \varepsilon)y_N, \quad B = x_N - (L + \varepsilon)y_N.$$

Since N is fixed, A and B are constants: they do not depend on n . Since $y_{n+1} \rightarrow \infty$, a constant divided by y_{n+1} tends to 0, so

$$\frac{A}{y_{n+1}} \rightarrow 0 \quad \text{and} \quad \frac{B}{y_{n+1}} \rightarrow 0.$$

Step 3: Choose the final threshold. Since $\frac{A}{y_{n+1}} \rightarrow 0$, there exists M_1 such that for $n \geq M_1$,

$$\left| \frac{x_N - (L - \varepsilon)y_N}{y_{n+1}} \right| < \varepsilon,$$

and since $\frac{B}{y_{n+1}} \rightarrow 0$, there exists M_2 such that for $n \geq M_2$,

$$\left| \frac{x_N - (L + \varepsilon)y_N}{y_{n+1}} \right| < \varepsilon.$$

Let $M = \max\{N, N_0, M_1, M_2\}$, where N_0 is the index from Step 1 with $y_{n+1} > 0$ for $n \geq N_0$.

For $n \geq M$, plugging these bounds into Step 1 gives

$$L - \varepsilon - \varepsilon < \frac{x_{n+1}}{y_{n+1}} < L + \varepsilon + \varepsilon,$$

that is,

$$L - 2\varepsilon < \frac{x_{n+1}}{y_{n+1}} < L + 2\varepsilon, \quad \text{so} \quad \left| \frac{x_{n+1}}{y_{n+1}} - L \right| < 2\varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, 2ε can be made as small as we like (given any target $\eta > 0$, run the argument with $\varepsilon = \eta/2$). Hence $\frac{x_{n+1}}{y_{n+1}} \rightarrow L$, and since shifting the index by one does not change the limit,

$$\lim_{n \rightarrow \infty} \frac{x_n}{y_n} = L.$$

□

Problem 3 (10 points)

Prove that if

$$\lim_{n \rightarrow \infty} a_n = L,$$

then

$$\lim_{n \rightarrow \infty} \frac{a_1 + a_2 + \cdots + a_n}{n}$$

exists and equals L .

Proof. Let

$$x_n = a_1 + a_2 + \cdots + a_n \quad \text{and} \quad y_n = n,$$

so that $\frac{x_n}{y_n} = \frac{a_1 + a_2 + \cdots + a_n}{n}$. We check the hypotheses of Problem 2.

The sequence (y_n) is increasing, since $y_{n+1} = n + 1 > n = y_n$. Also, $\lim_{n \rightarrow \infty} y_n = \lim_{n \rightarrow \infty} n = \infty$.

Next, since $x_{n+1} = a_1 + a_2 + \cdots + a_n + a_{n+1}$, we have $x_{n+1} - x_n = a_{n+1}$, and $y_{n+1} - y_n = (n + 1) - n = 1$. Thus

$$\frac{x_{n+1} - x_n}{y_{n+1} - y_n} = \frac{a_{n+1}}{1} = a_{n+1}.$$

Since $a_n \rightarrow L$, we also have $a_{n+1} \rightarrow L$, so $\lim_{n \rightarrow \infty} \frac{x_{n+1} - x_n}{y_{n+1} - y_n} = L$.

All hypotheses of Problem 2 hold, so $\lim_{n \rightarrow \infty} \frac{x_n}{y_n} = L$. Substituting $x_n = a_1 + \cdots + a_n$ and $y_n = n$,

$$\lim_{n \rightarrow \infty} \frac{a_1 + a_2 + \cdots + a_n}{n} = L.$$

□

Problem 4 (10 points)

Define the sequence $a_1 = 1$ and

$$a_{n+1} = 3 - \frac{1}{a_n}.$$

Show that a_n is convergent, by proving that a_n is monotone and bounded. What is the limit of a_n ?

$$\text{Let } \alpha = \frac{3 + \sqrt{5}}{2} \approx 2.618.$$

Claim. (a_n) is increasing ($a_{n+1} \geq a_n$) and $1 \leq a_n \leq \alpha$ for all n .

Proof. Step 1: Bounded. We prove $1 \leq a_n \leq \alpha$ by induction.

Base case. Since $a_1 = 1$, we have $1 \leq a_1 \leq \alpha$.

Inductive step. Assume $1 \leq a_n \leq \alpha$.

Since $a_n \geq 1 > 0$, we have $\frac{1}{a_n} \leq 1$, so

$$a_{n+1} = 3 - \frac{1}{a_n} \geq 3 - 1 = 2 \geq 1.$$

Since $0 < a_n \leq \alpha$, we have $\frac{1}{a_n} \geq \frac{1}{\alpha}$, and

$$\frac{1}{\alpha} = \frac{2}{3 + \sqrt{5}} = \frac{2(3 - \sqrt{5})}{9 - 5} = \frac{3 - \sqrt{5}}{2}.$$

Therefore

$$a_{n+1} = 3 - \frac{1}{a_n} \leq 3 - \frac{3 - \sqrt{5}}{2} = \frac{3 + \sqrt{5}}{2} = \alpha.$$

Hence $1 \leq a_{n+1} \leq \alpha$, and by induction $1 \leq a_n \leq \alpha$ for all n .

Step 2: Monotone. For every n ,

$$a_{n+1} - a_n = 3 - \frac{1}{a_n} - a_n = \frac{3a_n - 1 - a_n^2}{a_n} = \frac{-(a_n^2 - 3a_n + 1)}{a_n}.$$

The roots of $t^2 - 3t + 1 = 0$ are $\frac{3-\sqrt{5}}{2} \approx 0.38$ and $\frac{3+\sqrt{5}}{2} = \alpha$, so $t^2 - 3t + 1 \leq 0$ for $\frac{3-\sqrt{5}}{2} \leq t \leq \alpha$. By Step 1, $1 \leq a_n \leq \alpha$, and $\frac{3-\sqrt{5}}{2} < 1$, so a_n lies in this interval. Hence $a_n^2 - 3a_n + 1 \leq 0$, and since $a_n > 0$,

$$a_{n+1} - a_n = \frac{-(a_n^2 - 3a_n + 1)}{a_n} \geq 0.$$

Thus $a_{n+1} \geq a_n$ for all n , so (a_n) is increasing.

Step 3: Convergence. By Steps 1 and 2, (a_n) is increasing and bounded, so by the Monotone Convergence Theorem it converges. Let $a = \lim_{n \rightarrow \infty} a_n$.

Step 4: The limit. Since $a_n \geq 1$ for every n , taking the limit gives $a \geq 1$; in particular $a \neq 0$. Since also $a_{n+1} \rightarrow a$, letting $n \rightarrow \infty$ in $a_{n+1} = 3 - \frac{1}{a_n}$ gives

$$a = 3 - \frac{1}{a}.$$

Multiplying by a , $a^2 = 3a - 1$, i.e. $a^2 - 3a + 1 = 0$, so by the quadratic formula

$$a = \frac{3 \pm \sqrt{5}}{2}.$$

Since $\frac{3-\sqrt{5}}{2} < 1 \leq a$, the root $\frac{3-\sqrt{5}}{2}$ is impossible. Therefore

$$\lim_{n \rightarrow \infty} a_n = \frac{3 + \sqrt{5}}{2}.$$

□